

Muhammad Hamza

AI & WEB DEVELOPER

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PORTFOLIO portfolio-lime-rho-96.vercel.app

SUMMARY

Computer Science student and developer who has shipped 10 AI-powered, full-stack products end to end — from React/TypeScript interfaces to Python, Node and FastAPI back ends. My work spans retrieval-augmented generation (RAG), LLM-integrated features, real-time streaming apps and complete commerce and finance dashboards, most of them live and deployed. I learn by building — taking an idea from a blank repo to a working app in front of real users.

SELECTED PROJECTS

Folio — PDF Toolkit

React · TypeScript · Node · Express

A full-stack, ad-free PDF toolkit with 19 tools — merge, split, compress, watermark and Office ⇌ PDF conversion — served from one Docker container, with files deleted after download.

muhammadhamza221003-folio.hf.space · github.com/MuhammadHamza2210/Folio

Synapse — AI Study OS

React · FastAPI · RAG · Three.js

A retrieval-augmented “Study OS” that ingests a user’s own documents into a vector store and answers questions grounded in their material, with recalled concepts shown as a 3D Mind Palace.

synapse-eight-ruby.vercel.app · github.com/MuhammadHamza2210/Synapse

CampusFind — Campus Marketplace & Lost-and-Found

React · Node · MongoDB · Socket.io

A full-stack platform for university-verified students to post lost, found or for-sale items — with automatic lost↔found matching, ownership-verified claims and real-time chat over WebSockets.

campus-find-mu.vercel.app · github.com/MuhammadHamza2210/CampusFind

CodeScan AI — Code Plagiarism Detector

Python · ML · CodeBERT

Combines four engines — AST diffing, a trained ML model, winnowing fingerprints and CodeBERT embeddings — for structure-aware similarity scoring that resists renaming and reformatting.

github.com/MuhammadHamza2210/CodeScan-AI

Also built: Orion (real-time stock dashboard, WebSockets + AI) · Fluxx (offline-first finance dashboard, React/IndexedDB) · WildMind AI (Gemini-powered wildlife encyclopedia) · LootLo (full-stack e-commerce with admin console, Flask/SQLite) · Hamza Restaurant (Node/Express ordering system, JWT auth) · WeatherScope (Python data viz).

SKILLS

Languages Python · TypeScript · JavaScript · SQL

AI / ML RAG & LLM apps · Prompt engineering · LLM APIs (Gemini, Claude, Ollama) · embeddings & vector search

Frontend React · Vite · Tailwind CSS · Three.js / R3F · Framer Motion

Backend Node.js · Express · FastAPI · Flask · REST APIs · WebSockets

Tools Git & GitHub · SQLite · Streamlit · Vercel

EDUCATION

BS Computer Science — Bahria University, Karachi

2024 – 2027 (expected)

CGPA **3.81 / 4.0** — through 5th semester

Relevant coursework: Data Structures & Algorithms · Databases · OOP · Software Engineering · Operating Systems